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Relations between the development of school investment, self-confidence, and language achievement in elementary education: a multivariate latent growth curve approach

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Abstract

Latent growth curve (LGC) analysis of longitudinal data for pupils' school investment, self-confidence and language ability is presented. A multivariate model is tested that relates the three developmental processes to each other and to intelligence. All processes show significant differences between children in their developmental curves. The increase in language ability and the decrease in school investment correspond with the hypotheses. No hypothesis is formulated about self-confidence, but an increase for some and a decrease for others are found. The hypothesis that development in language ability, school investment, and self-confidence are mutually positively associated is supported, as is the hypothesis that intelligence accounts for some of the differences in language ability. School investment, self-confidence, and intelligence each explain a different part of development in language ability.
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1. Introduction

Children differ in the knowledge they already have at the start of elementary education, as well as in how their achievement develops during elementary education. Academic achieve-

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ment starts to become important in elementary school as pupils are prepared for secondary education. In the Netherlands, secondary education is divided into quite different academic achievement tracks, varying from schools preparing for lower vocational levels to schools preparing for university. Consequently, factors influencing development of achievement in elementary education seem quite crucial for pupils' educational careers, and not only in the Netherlands.

Motivation and emotion with respect to learning are generally regarded as contributing to differences in academic achievement, on top of characteristics influencing achievement, such as intelligence or earlier achieved ability (see e.g., Kurdek & Sinclair, 2001; Vroom, 1964). In the European context, research on motivation for learning has usually addressed students' motivation in secondary education. Problems in school investment have also more often been found in secondary education than in elementary education, while an earlier onset is quite possible. Though, in American motivation research, younger learners (from the higher grades of elementary education) are often included, research incorporating the earliest years of education is less common. Differences in pupils' development in achievement, motivation, and emotion towards school might already be present from the start. However, it is difficult to measure the concepts directly at young children at the start of elementary education.

Standard pupils' achievements, for instance in language, are developing during elementary education. Some pupils develop well and others less than average. Questions arise as to what extent the development of achievement is related to pupil's motivational and emotional development. Longitudinal studies that also cover the earliest years of education are necessary to answer questions like these. To be more explicit, as noted by Pintrich (2000), longitudinal research in the area of motivation should adopt an individual growth curve perspective. Indeed, growth curve modeling is well suited to answer questions concerning systematic differences in individual developmental processes.

With respect to developments in students' motivation, a general decrease during secondary education has been found repeatedly (Eccles & Midgley, 1989; Peetsma, 1997; Vorst, 1985). Eccles and Midgley (1989) explain the decrease in motivation for school with a lack of 'person–environment fit'. In secondary education, a misfit between school life and the students' psychological development in adolescence keeps growing. Peetsma (1997) shows students' time perspective on school becomes less important than perspectives on other domains of life, such as personal development. It seems that the perspectives on other domains of life become motivationally concurring with the perspective on school and the subsequent professional career. A decrease in motivation during higher education has also been found (e.g., Hofman et al., 2001). Decrease in school motivation does not seem to show up only in adolescence, but may well be inherent to any tier in education. It might, thus, be expected in elementary education as well.

The concept of motivation may seem unambiguous in everyday language; in scientific theory it is a concept in which many aspects can be distinguished (see, e.g., Boekaerts, 2001). At elementary school age and certainly in kindergarten many aspects of students' motivators for learning, for instance, different goal orientations, will not be active or hardly distinguishable. But from the start of elementary education, children will already put more or less

investment in school, which can be observed by their teachers. With respect to investment in school, the aspects 'intensity', 'perseverance', 'continuing motivation', and 'direction' of students' working behavior in school are distinguished (Maehr & Braskamp, 1986; Roede, 1989). Of these, intensity and perseverance are probably the most relevant in elementary schools. The more personal investment, intensity and perseverance in working on a school task, the better achievements can be expected (Peetsma, 1992).

Besides motivation and intelligence or ability, children's emotional well-being and affections in the school situation are related to school achievements (Boekaerts, 1993). Emotional well-being or feeling confident or in control of the school situation seems important for motivation and learning (Peetsma, 2000; Seiffert & O'Keefe, 2001). Mavrogenes and Bezruczko (1993) found a positive influence of self-confidence on writing development for pupils at risk.

Students' self-confidence proved to be more strongly related to achievement than other affective variables reported by Fennema (1984) and Meyer and Fennema (1986). Fennema and Sherman (1977) found that students with high mathematics self-confidence had low mathematics anxiety and vice versa, and Meijer (2001) detected a positive relation between anxiety and self-confidence. It seems that self-confidence and anxiety function as opposite affections.

Self-confidence can be hard to measure at an early age. Opposite emotions to feelings of self-confidence, generating being anxious or upset in the school situation, will be observed by their teachers. The observation of being easily anxious or upset can function as a proxy of self-confidence in the school situation. In this, we expect developmental processes in self-confidence to be related positively to developments in investment in school and achievement.

Although, children's intelligence is generally assumed to be important for achievement, IQ scores have not often been included in research on achievement. These scores, obtained at an age when they can be assumed to be an acceptably reliable measure of intelligence, are expected to be positively related to achievement.

1.1. Research questions

The aim of this study is to investigate the relationship between the development of pupils' language achievement, school investment, and self-confidence during elementary education. The main research questions and hypotheses are:

1. How do pupils' school investment, self-confidence, and language achievements develop in elementary education (from kindergarten to secondary education)? We expect an increase in language achievement, but we expect decreasing school investment during elementary education. With respect to self-confidence, we have no expectation on the direction of development during elementary education.
2. Is the developmental process of language achievement in elementary education related to school investment and self-confidence? We expect developments in language achievement, school investment, and self-confidence to be mutually positively associated. With respect

to school investment, this implies the more positive the developments in achievement and self-confidence, the less the decrease in school investment.

3. To what extent is intelligence related to developmental processes in school investment, self-confidence, and language achievements in elementary education? We expect that though some of the individual differences in development of language ability can be attributed to intelligence, enough differences remain to be attributed to school investment and self-confidence, so the relations between the variables will still exist after including IQ in the analyses.

2. Method

2.1. Subjects

Data on the subjects were taken from the longitudinal PRIMA cohort study in The Netherlands (Driessen, Langen, & Vierke, 2000; Jungbluth & Meijnen, 1994). In this research project, about 600 elementary schools were sampled for the first time in 1994. Every 2 years children from Grades 2, 4, 6, and 8 are tested in language and rated by their teachers on a number of attitudinal and motivational aspects. The fourth measurement occasion of this study took place in the 2000/2001 school year, creating the opportunity to study the development of children throughout elementary education: beginning at the second grade (kindergarten: 5–6 years of age) in school year 1994/1995 up to the eighth grade (11–12 years of age) at the last measurement. The first measurement, at the end of kindergarten, can be considered as the starting point for the more structured part of elementary education.

As in most longitudinal studies, the PRIMA project suffers from dropout of both schools and children. Although the sample is supplemented with new schools at every measurement, the number of schools and children available for longitudinal analyses decreases with every successive measurement occasion in the project. The loss of schools has been shown to be random: the schools lost do not differ in any essential way from schools that continue to participate in the project (Roeleveld & Portengen, 1998). The loss of individual children within participating schools, however, is not completely random. Although most loss has to do with parents moving, other reasons are transfer of children to special education or staying down a grade. As a result, children dropping out of the study score, in general, are somewhat lower on the cognitive tests. After four measurement occasions, we have at our disposal longitudinal data on 2693 pupils in 214 schools. And although there is some selectivity in this sample, it is still representative for the vast majority of children that follows a "normal," i.e., nondelayed, educational career in elementary schools in The Netherlands.

2.2. Instruments

Our main interest in this study is the development of achievement, especially the language achievement of the children. Since this achievement grows considerably between the second

and the eighth grade, a different test had to be used at each grade. The language tests were especially developed for the PRIMA project by CITO, the educational testing and measurement company of the Netherlands (Driessen, Langen, & Oudenhoven, 1994). Each test was made up of about 60 morphological, syntactical, and semantic items and was intended to measure basic language skills that can be seen as prerequisites for all school achievement. Reliability scores for the tests were good, varying from .85 to .96. To compare the language achievement of children over time, test scores were calibrated using OPLM-procedures from item–response theory (Verhelst, Glas, & Verstralen, 1993; Vierke, 1995). As a result, we have scores of the children on a continuing one-dimensional scale for language ability on each of the four measurement occasions.

Because of the young age of the children, no direct measurements of self-confidence and school investment were performed in the PRIMA project. As a proxy, we use indicators taken from the Pupils Profile (Jungbluth, Langen, Peetsma, & Vierke, 1996), which was filled out by the teachers. In this instrument, a list of items concerning children's behavior, attitudes, and motivation had to be scored on a five-point Likert-type scale. Unfortunately, the instrument went through some changes over the four waves in the PRIMA project (Jungbluth, Roede, & Roeleveld, 2001). Thus, in order to have identical measurements of self-confidence and motivation over the years, we can only use items that have been unchanged.

During the first three measurement occasions, teachers were asked to fill out the Pupils Profile for all children in their class. This proved to be an onerous task, which resulted in responses for about 80% of the children. To meet teachers' complaints about the amount of work, they only had to complete the Profile for a randomly selected half of their pupils at the last measurement occasion. So at this last measurement, we have only about 50% of the teacher ratings for self-confidence and motivation, but missing values can be considered to be completely random.

For self-confidence, two negatively scaled items were used as indicators at all time points: 'this child is anxious and frightened' and 'this child is easily upset'. The items refer, in particular, to a general opposite affective character of self-confidence. Investment in school was measured by one positively scaled item: 'this child works accurately' and by one negatively scaled item: 'this child easily thinks work is done'. These items represent two key aspects, intensity and perseverance, of school investment. As indicators for the intelligence of the children, we used scores on two nonverbal tests (Figures and Exclusion), which were administered in Grade 6, when the children were 9–10 years of age. These tests were developed for the evaluation of educational policies in The Netherlands, using items from several standard IQ tests (Doddema-Winsemius & Werf, 1989). They have good reliability scores (about .75) and show sufficient validity (Driessen et al., 1994).

2.3. Modeling strategy

Latent growth curve (LGC) analysis was used to analyze the data. LGC analysis represents repeated measures of a given concept as a function of time and other measures

(Willett & Sayer, 1994). An LGC analysis specifies an individual growth curve for each subject to represent the development over time. All subjects in a given population are assumed to have developmental curves of the same functional form (e.g., all linear), but the parameters describing their curves may differ. With linear developmental curves, for example, individual differences may be due to heterogeneity in the initial level, as well as heterogeneity in the growth rate or rate of change. In other words, the subjects differ in their level at the first measurement occasion and develop at different rates. The initial level and the growth rate will be referred to as the 'growth parameters'; that is, the parameters describing the pattern of growth of each individual child. In the current research, the focus is on the intra- and interindividual variation of the three repeatedly measured variables (language ability, self-confidence, and school investment), in particular, with regard to the covariation across individuals with respect to the patterns of development on these variables (cf. MacCallum, Kim, Malarkey, & Kiecolt-Glaser, 1997). We are interested, for example, in the relation between the developmental process of self-confidence and the developmental process of language ability. This type of analysis is called a multivariate growth curve analysis. A brief description of LGC analysis is presented below; a detailed description of the approach is beyond the scope of this article (we refer to Willett & Sayer, 1994 for a description of the univariate growth curve analysis; to Duncan, Duncan, Strycker, Li, & Alpert, 1999; MacCallum et al., 1997; Muthén & Khoo, 1998 for a detailed explanation of both univariate and multivariate growth curve models; and to Chan, Ramey, Ramey, & Schmitt, 2000 for a recent application of a multivariate growth curve model).

It is convenient to view the initial level and the growth rate as latent variables (Muthén & Khoo, 1998). Conventional structural equation modeling software can then be used to analyze the models, and path diagrams can be used to give a graphic presentation. Fig. 1 gives such a graphic presentation of a growth model. It does not show the individual growth

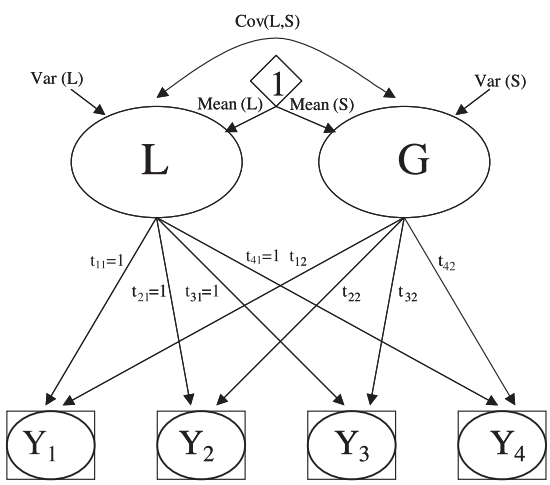

Fig. 1. Graphic representation of LGC model.

curves explicitly, but it shows the means and variances of the parameters describing the growth curves, and their correlation. The variables denoted Y_1 to Y_4 represent the consecutive measurements of the outcome. The symbol of a circle inside a square represents observed variables that have been measured for certain participants but not for others (following Ghisletta & McArdle, 2001; McArdle, 1994). The ellipses represent the latent variables, *L*(evel) and *G*(rowth rate), respectively, and contain the means (regression on the diamond) and variances of the (initial) level and the growth rate of the curves, respectively. They provide information on the interindividual differences in the developmental curves, and are allowed to correlate (e.g., a positive correlation between the initial level of language ability and growth implies that children with a higher initial level have a larger growth rate than children with a lower initial level). However, substantive interpretations of this parameter can only be made if the origin of the developmental process corresponds to the first measurement occasion (Stoel & Van den Wittenboer, in press). In the present research, the initial level corresponds to the score of the children at kindergarten (first occasion). This may be reasonable here, since the last year of kindergarten can be regarded as the start of any school career. However, of primary interest is the growth rate, which corresponds to development over the subsequent years (second to fourth occasion). Nevertheless, with some caution, we will also include inferences regarding the level parameters in our results and discussion.

Fig. 1 actually is a confirmatory factor model, with the difference that the factor loadings are constrained to specific values. The factor loadings for L=[t_{11}, t_{21}, t_{31}, t_{41}] are fixed to 1. If the factor loadings for G=[t_{12}, t_{22}, t_{32}, t_{42}] are constrained to [0, 2, 4, 6], respectively, this implies linear growth, where the interpretation of G is the change per year. Following Meredith and Tisak (1990), two factor loadings for G need to be constrained in the nonlinear growth curve model (e.g., [0, t_{22}, t_{32}, 6] following McArdle & Bell, 2000). L now represents the level at the first measurement occasion and G represents the mean growth per year over the 6 years. In the linear model, growth is characterized by a constant increment, whereas in the nonlinear model increments are not equal. If, for example, the factor loadings are estimated as [0, 3, 5, 6], this implies that growth is strongest between the second and the first occasion, and that it gets less strong during the following occasions.

The motivational variables are measured by their raw score indicators. This allows them to be modeled as latent constructs in a longitudinal factor model, and to test them for the assumption of measurement invariance. Measurement invariance (the invariance of the measurement parameters) ensures a comparable definition of the latent construct over time (Hancock, Kuo, & Lawrence, 2001). Violation of the assumption of measurement invariance renders it impossible to assess change within a subject, because this change will be confounded with the change in meaning of a construct over time. Measurement invariance can exist to a certain degree. Though some authors require full measurement invariance in testing hypotheses concerning repeatedly measured latent constructs (Oort, 2001), others have argued (Byrne, Shavelson, & Muthén, 1989; Pentz & Chou, 1994) that some violations may be admissible. Partial measurement is defined, then, by a measurement model that includes a combination of measurement parameters (factor loadings and

intercepts), some of which are free and others constrained to be equal over time (Pentz & Chou, 1994).

The measurements of language ability, calibrated with item–response theory, resulted in one score for each child at each measurement occasion. An additional test for measurement invariance of this variable could not, therefore, be performed. For intelligence, it is assumed that each indicator measured intelligence equally well. This implies equal factor loadings for both subtests 'Exclusion' and 'Figures'.

For the models reported in this article, we used Mplus 1.04 (Muthén & Muthén, 1998) to fit the models to the data. Both means and covariances were analyzed simultaneously. Thus, unlike usual structural equation models, the factor means are not assumed to be zero. The chi-square measure of overall goodness of fit (CHISQ), in combination with the root mean square error of approximation (RMSEA), was used to evaluate the overall goodness of fit of the models. RMSEA is added since the chi-square measure is not always appropriate to evaluate structural equation models (Browne & Cudeck, 1992). They propose RMSEA values smaller than .05 to be indicative of a close fit, and values smaller than .08 to be indicative of a reasonable fit. For the comparison of competing models (e.g., nonlinear vs. linear), the chi-square difference test (ΔCHISQ) can be used.

Our sample has a considerable amount of missing data, but most of these are missing 'by design'. As explained in the 'data' section, the missing data can be considered to be completely random (MCAR; Little & Rubin, 1987). The present research study uses the full information maximum likelihood (FIML) method of Muthén, Kaplan, and Hollis (1987), as implemented in the software package Mplus 1.04. This method draws on the missing data theory of Little and Rubin (1987), and utilizes the raw data as input, instead of the observed covariance matrix and means vector. In a recent simulation study (Enders & Bandalos, 2001), this method was found to give better estimates than naive methods, such as pairwise or listwise deletion in the presence of missing data. The method has also been shown to perform well in combination with LGC models (Wothke, 2000).

In constructing our final model, the strategy advocated by Andersen and Gerbing (1988) has been applied (see Garst, Frese, & Molenaar, 2000, for a recent application). This strategy consists of two parts: (1) testing the measurement models, and (2) combining the measurement models in a structural model. The parts can further be subdivided into several steps. The main rationale behind this strategy is that possible model misspecifications can be located more easily, and necessary additional tests can be performed more adequately. A good strategy is important, as the validity of the final model depends on the validity of the parts from which it is constructed.

3. Results

3.1. Measurement models

The estimated covariance matrix and means vector is presented in Appendix A. Each repeatedly measured variable is named in terms of its underlying construct, the measurement

occasion (grade), and, if necessary, its number. For example, 'inv21' represents the first indicator of school investment at Grade 2.

The measurement models of school investment and self-confidence were found to follow a similar pattern. For each model, the factor loadings of the indicators could be constrained to be equal over time, and the intercepts at all but the first occasion, besides the necessary constraints for identification purposes. Residuals of the same indicators were correlated across time up to two occasions. The measurement models both have good overall goodness of fit measures. The chi-square measure of fit is nonsignificant (α=.01) and the RMSEA < .05. Self-confidence has a fit of: CHISQ(9) = 17.18, P=.05; RMSEA=.018, and school investment: CHISQ(9) = 8.97, P=.44; RMSEA=.000. We may conclude that partial measurement invariance has been found for both models, the

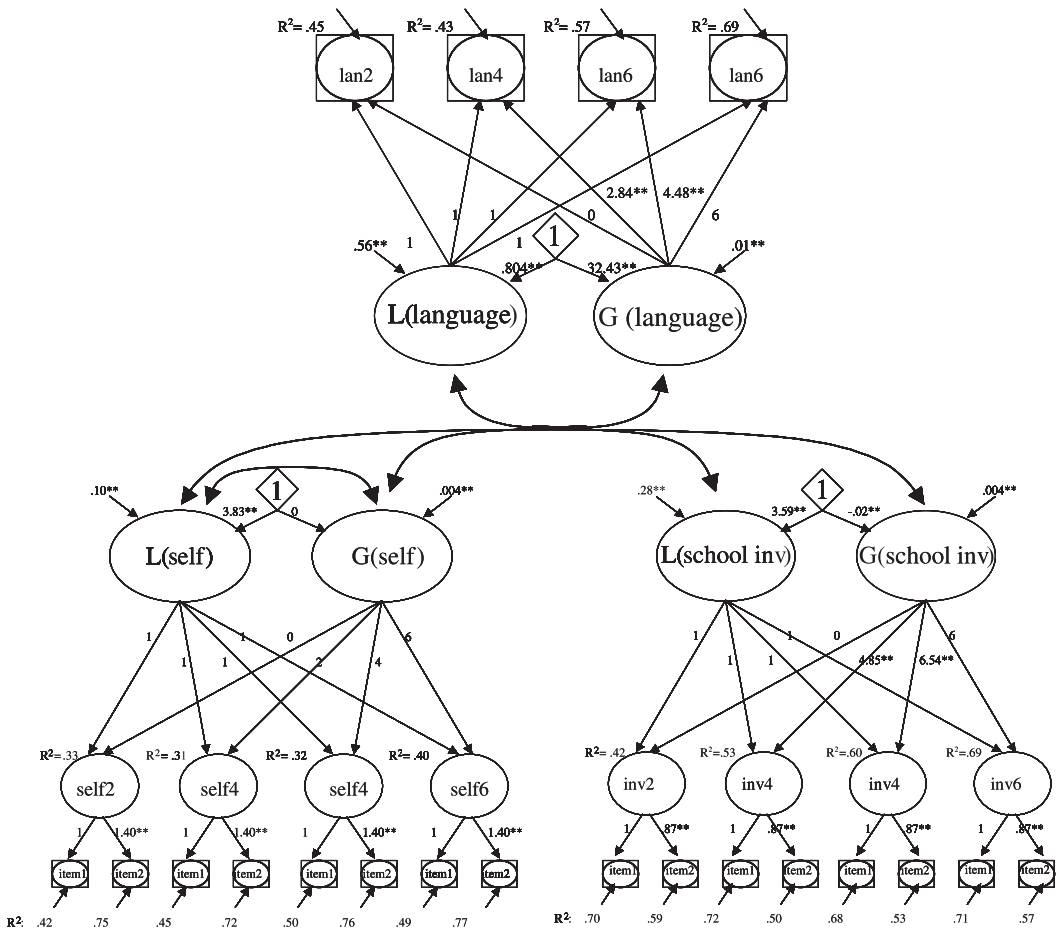

Fig. 2. Multivariate growth curve model of language ability, self-confidence, and school investment. lan = language ability, self = self-confidence, inv = school investment.

parameter estimates of which are presented in the multivariate growth model of Fig. 2. We may summarize that children's school investment and self-confidence as indicated by judgments from different teachers at each measurement occasion show partial measurement invariance. This supports the validity of the method in which professional teachers judge pupil's noncognitive characteristics. Due to the satisfactory measurement models obtained, one can reasonably assume that the same construct has been measured over time.

3.2. Univariate growth curve models

From the hypotheses, developmental processes can be deduced for the three concepts. However, specific hypotheses about the functional form, or shape, of the growth processes have not been formulated. For this reason, we started with nonlinear models (Meredith & Tisak, 1990). If the resulting model goodness of fit was good, a further test could be performed for linearity by constraining the appropriate parameters. If these constraints did not lead to a serious deterioration in model fit, it was assumed that the developmental process could be modeled with a linear growth rate.

Table 1 presents the goodness-of-fit measures of the univariate growth curve models for the three concepts under investigation. For self-confidence, the nonlinear model (Model 1) provided a good fit; we could therefore continue posing constraints resulting in a linear model (Model 1.1). The chi-square difference test indicated that the linear model did not fit significantly worse (ΔCHISQ = 1.46, $\Delta df = 2$). Inspection of the parameter estimates led to a further constraint. The mean of the growth rate did not differ significantly from zero, and could therefore explicitly be constrained to zero (Model 1.2). The chi-square difference between the resulting Model 1.2 and Model 1.1 was not significant (ΔCHISQ = 1.77, $df = 1$). The model could not be constrained any further. The final Model 1.2 for self-confidence provided the following goodness-of-fit measures: CHISQ(15)= 27.27, P=.03, RMSEA=.018. The parameter estimates of this and the following univariate growth models are discussed in the Multivariate growth curve model section.

The nonlinear Model 2 for school investment also resulted in a good fit. The linear Model 2.1, however, fitted significantly worse (ΔCHISQ = 92.59, $\Delta df = 2$), and the assumption of linear growth had to be rejected. Inspection of the parameter estimates of the nonlinear model indicated that the covariance between the initial level and the growth rate could be fixed to zero (Model 2.2). This constraint did not lead to a serious deterioration in model goodness of fit (ΔCHISQ = 0.12, $\Delta df = 1$). The final Model 2.2 for school investment provided the following goodness-of-fit measures: CHISQ(13)= 14.72, P=.33, RMSEA=.007.

The results for language ability (Model 3) were essentially the same as for school investment. The nonlinear model provided a reasonably good fit. Although the CHISQ was significant, the RMSEA was below .05 and the 90% confidence interval did not exceed 0.08. A linear constraint (Model 3.1) on the growth curves led to a significant decrease in model goodness of fit (ΔCHISQ = 1041.19, $\Delta df = 2$), and we had to continue with the nonlinear model. The covariance between the initial level and the growth rate could be

Table 1
Goodness-of-fit measures of the univariate growth curve models

Model		*df*	CHISQ (*P*)	RMSEA
Self-confidence				
1	nonlinear growth	12	24.04 (.02)	.019 (.007–.031)
1.1	linear growth	14	25.50 (.03)	.018 (.005–.028)
	difference 1.1 and 1	2	1.46	
1.2	mean growth = 0	15	27.27 (.03)	.018 (.006–.028)
	difference 1.2 and 1.1	1	1.77	
School investment				
2	nonlinear growth	12	14.60 (.26)	.009 (.000–.023)
2.1	linear growth	14	107.19 (.00)	.050 (.041–.059)
	difference 2.1 and 2	2	92.59**	
2.2	corr(level, growth) = 0	13	14.72 (.33)	.007 (.000–.021)
	difference 2.2 and 2	1	0.12	
Language ability				
3	nonlinear growth	3	20.81 (.00)	.047 (.029–.067)
3.1	linear growth	5	1062.0 (.00)	.280 (.266–.294)
	difference 3.1 and 3	2	1041.19**	
3.2	corr(level,growth) = 0	4	21.53 (.00)	.040 (.025–.058)
	difference 3.2 and 3	1	0.72	

The 90% confidence interval for RMSEA is given in parentheses. $N_{total} = 2693$ for language ability; $N_{total} = 2660$ for self-confidence and school investment. The difference in total sample size is due to the fact that a subsample of $n = 33$ subjects have no observations for self-confidence and school investment.

** $P < .01$.

restricted to zero (ΔCHISQ = 0.72, $\Delta df = 1$). Model 3.2 provided the following goodness-of-fit measures: CHISQ(4)= 21.53, P=.00, RMSEA=.040.

The results, therefore, indicate a well-fitting linear growth curve model for the concept of self-confidence, and a nonlinear growth model for both school investment and language ability. In the next section, these three models are combined to get information regarding the relations between them. To secure measurement invariance, and the functional form of the growth curves in the multivariate model, the appropriate parameters are constrained to the values as estimated in the univariate models.[1]

[1] The factor loadings of the univariate growth curve models were fixed to the estimated values from the univariate analysis. This was done to prevent parameter changes, and to keep the interpretation of the multivariate growth curve model in agreement with the univariate models. To test the fit of the multivariate model, the degrees of freedom have to be corrected by subtracting the number of a priori fixed parameters, so that the values of the RMSEA fit measure were recalculated according to

$$RMSEA_{new} = \sqrt{\frac{\mathrm{RMSEA}^2_{\mathrm{old}}}{df_{\mathrm{new}}} df_{\mathrm{old}}}$$

3.3. Multivariate growth curve model

The main rationale of this paper is to investigate the relationships between growth processes in language ability, school investment, and self-confidence. We will now, therefore, report the results of the integration of the univariate growth curve models into the multivariate growth curve model. The focus will be on the relations between the levels and growth rates of the three processes, before and after the effect of IQ has been controlled for. The parameter estimates of the complete multivariate model are shown in Fig. 2. Table 2 presents the correlations between the growth parameters of the three processes in the multivariate growth curve model. Before the correlations between the growth parameters are described, we will describe the parameter estimates of the three growth processes.

The multivariate growth curve model provided the following goodness-of-fit measures: CHISQ(141)= 290.83, P=.00, RMSEA=.020. In this model, some of the correlations between the growth parameters did not differ significantly from zero. For reasons of parsimony, these correlations are explicitly fixed to zero, resulting in the following goodness-of-fit measures: CHISQ(146)= 292.86, P=.00, RMSEA=.019. The chi-square difference test indicated that this model did not fit significantly worse (ΔCHISQ = 0.03, $\Delta df = 5$). For all three processes, the associated R^2 values are substantial (ranging from .31 to .69) indicating that a substantial proportion of the variance was accounted for by the growth factors. The concurrent correlations between the first-order factors of self-confidence and school investment were estimated; their values were .18**, .11**, .11**, and .11**, respectively. The parameter estimates for the separate growth processes can be interpreted as follows. The mean linear growth curve for self-confidence has an initial level of 3.83 and a growth rate of zero. The variance of the initial level and the growth rate are significant ($P < .01$), indicating individual differences between the children in the development of self-confidence. In other words, the children have different growth curves. The amount of self-confidence increases for some children, while it decreases for others. The correlation between the initial level and the growth rate of $r = -.39$** indicates a negative association, so a lower start corresponds to a higher growth rate, and vice versa.

Table 2
Correlations between growth parameters in the multivariate growth curve model

	1.	2.	3.	4.	5.	6.
1. Level (language)	1					
2. Growth (language)	0	1				
3. Level (self-confidence)	.23**	−.18**	1			
4. Growth (self-confidence)	0	.49**	−.39**	1		
5. Level (school investment)	.42**	.16*	0	.30**	1	
6. Growth (school investment)	0	.26**	0	0	0	1

* $P < .05$.
** $P < .01$.

The mean nonlinear growth curve for school investment has an initial level of 3.59 and a growth rate of -0.02, both having a significant variance ($P<.01$). The correlation between the initial level and the growth rate is zero, so there is no association between the initial level of school investment and growth. Note that the factor loadings of the second and third occasion were estimated, respectively, as 4.85 and 6.54 in the prior analysis. When these factor loadings, [0, 4.85, 6.54, 6], are contrasted with the time points of the measurement occasions, [0, 2, 4, 6], the nonlinearity pops up as illustrated in Fig. 3a. The decrease in school investment is followed by an increase at the last measurement occasion. As can be seen in Figs. 2 and 3a, the mean of the growth rate factor is relatively

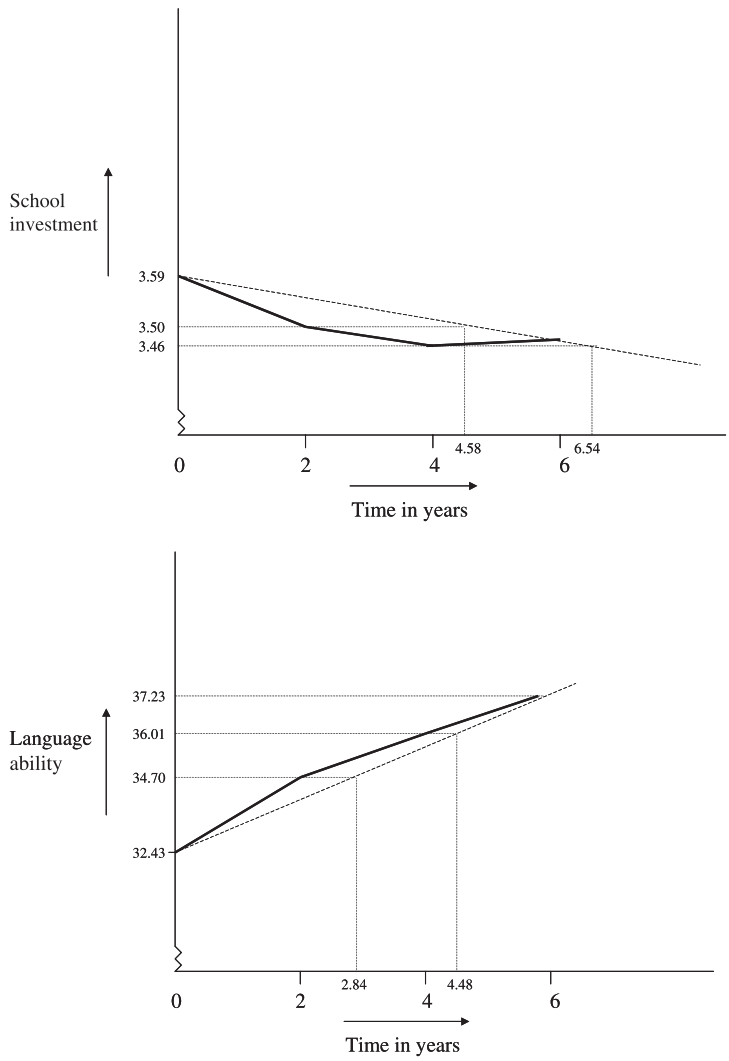

Fig. 3. Mean LGC for school investment and language ability.

small. It did differ significantly from zero, however. Though small, and not significant, the decrease in school investment corresponds with our expectations.

The mean growth curve for language ability has an initial level of 32.43 and a growth rate of 0.80, with significant variance. The correlation between the initial status and the growth rate is zero. The factor loadings of the second and third occasion are estimated as 2.84 and 4.48, respectively. As one can see in Fig. 3b, the growth in language ability gets smaller at the consecutive measurement occasions.

Summarizing these results, we can conclude that the individual developmental processes of self-confidence, school investment, and language ability show substantial interindividual differences, in both the level at the first measurement occasion and subsequent growth during the following years of education. In the next paragraphs, we will describe the relations between the growth parameters in order to investigate the association between the growth processes within each individual child.

As can be seen from Table 2, development in school investment and self-confidence show a positive correlation with development in language ability. The relatively high correlations of growth in language ability with growth in school investment and self-confidence are r=.26** and r=.49**, respectively. This indicates that children with a higher growth in language ability show a smaller decrease in school investment and a higher increase in self-confidence, and vice versa. The zero correlation indicates that no relation is found between the development in school investment and self-confidence.

There is one significant correlation between the growth processes in self-confidence and school investment. Only the correlation between the initial level of school investment and the growth rate for self-confidence is significant, r=.30**. In other words, children starting with a higher level of school investment show a greater increase in self-confidence. Furthermore, the correlations of both levels of self-confidence and school investment with level of language ability are relatively high (r=.23** and r=.42**, respectively). This points to a positive association between them. Children starting with higher language ability, also have higher self-confidence and school investment at that time. The growth in language ability shows a small, though significant, negative association with the initial level of self-confidence ($r = -.18$**), and a small, though significant, positive association with the initial level of school investment (r=.16*).

The fit of the multivariate growth curve model with intelligence is also reasonably good: CHISQ(180) = 360.49, P=.00, RMSEA=.018. The correlations between the growth parameters while controlling for the influence of intelligence are presented in Table 3. The first row of Table 3 presents the estimated standardized coefficients of the regression of the growth parameters on intelligence. As can be seen, intelligence has a considerable effect on the initial levels of language ability (.54**) and school investment (.47**); and a smaller, though significant, effect on the growth rate of language ability (.28**) and self-confidence (.23**). In other words, some of the differences in these parameters between the children are related to intelligence. Next, comparing the correlations of the growth parameters before and after the effect of intelligence has been controlled for, the effect of intelligence on association between the growth parameters can be investigated.

Table 3
Regression coefficients of the growth parameters on intelligence, and correlations between growth parameters in the multivariate growth curve model controlled for intelligence

	1.	2.	3.	4.	5.	6.
Growth parameters on intelligence	.54**	.28**	.09	.23**	.47**	.02
1. Level (language)	1					
2. Growth (language)	0	1				
3. Level (self-confidence)	.12**	−.15**	1			
4. Growth (self-confidence)	0	.35**	−.42**	1		
5. Level (school investment)	.18**	.06	.0	.14**	1	
6. Growth (school investment)	0	.27**	0	0	0	1

** $P<.01$.

The decrease in the correlation of initial level of self-confidence and school investment with the initial level of language ability to r=.12** and r=.18**, respectively, indicates that intelligence accounts for some of the association between these parameters. The same holds for the correlation between the growth in language ability, growth in self-confidence (r=.35**), and the initial level of school investment (r=.06). Intelligence accounts for almost all the association between the initial level of school investment with the growth of language ability; this correlation is not significantly different from zero. The association between the growth in language ability and the growth in school investment remains more or less the same.

In accordance with our expectations, therefore, development in both school investment and self-confidence is positively related to development in language ability. Intelligence has an effect on development in language ability, and accounts for some, but certainly not all, of the association between the processes.

4. Conclusion and discussion

The topic of our research was to investigate the development in language ability, school investment, and self-confidence in relation to each other. First, the results indicate that the developmental processes of school investment, self-confidence, and language ability show considerable individual differences; that is, the development of the children follows different trajectories. For self-confidence, the trajectory seems to be linearly related to time. An increase for some of the children is contrasted by a decrease for others. Development in school investment and language ability was found to be nonlinearly related to time: overall school investment decreased slightly while language ability increased, as expected. These results support the expected decrease in school investment over the period of attendance at elementary school, except for the small increase in school investment at the end of the school period.

Development in language ability, school investment, and self-confidence were, as expected, mutually positively associated. Children showing a greater increase in their

language ability also showed a greater increase in their self-confidence, as well as a smaller decrease in their school investment. These results are in accordance with our second hypothesis. The multivariate growth curve model was estimated both before and after the effect of intelligence had been controlled for. Doing this made it possible first to assess the effect of intelligence on the developmental processes directly, and second on the relationships between the parameters describing these developmental processes. Regarding the first point, it was found that intelligence accounts for some of the individual differences in the development of language ability, school investment, and self-confidence. To be more explicit, the more intelligent the children, the higher is their level at the start, and the more positive is their development on all three concepts. Regarding the second point, it was found that intelligence accounts for some, but not all, of the association between the developmental processes. Though some of the individual differences in the development of language ability can be attributed to intelligence, it does not explain all. There remain enough differences that may be attributed to school investment and self-confidence. The fact that the processes of school investment and self-confidence are relatively unrelated supports the argument that school investment, self-confidence, and intelligence each may explain a different portion of the individual differences in the development of language ability. This result offers confirmation for the third hypothesis.

A point of discussion may be the role of the initial level of the three processes. Although it is tempting to give substantive interpretations to these parameters, care should be taken in doing so. An initial level implies a starting point of development; that is, before this point no development existed. Therefore, strictly speaking, a substantive interpretation of the initial level is only possible if the first measurement occasion corresponds to the origin of the developmental process. In this research, the first measurement occasion was in kindergarten (Grade 2) before the start of structured education in Grade 3. Therefore, it can be argued that language ability, self-confidence, and school investment for school as measured at the first occasion represents the level before any structured education was received, and so can be regarded as being the initial level. From this point of view, substantive conclusions may be based on these parameters.

Although longitudinal analyses were performed, one should take care in giving a causal interpretation to our results. The relationships between the growth parameters of the three processes cannot all be convincingly interpreted causally. As mentioned by, for example, Garst et al. (2000), a prerequisite for a causal interpretation of a given phenomenon is the time order effect. For example, the correlations between the growth rates give up the time order because they look at general trends over the full time range (Garst et al., 2000). Though of minor interest, the correlations between the initial level and the growth rate could more safely be given a causal interpretation. An initial status comes logically before any development. If the influence of third variables could be ruled out, then a correlation may be interpreted as an effect of the initial level on the growth rate (Stoel & Van den Wittenboer, in press).

At the start of education, the relation between ability and self-confidence still has to be established (see, e.g., Skaalvik & Valas, 1999), consequently, pupils' self-confidence will

become more important during elementary education. We can have some doubts about the zero point just before structured education starts. As Nicholls (1978) pointed out, it is only at about the age of 9 years that almost all children do clearly distinguish between ability, not in their control, and effort, which is mainly in their own control. Therefore, further investigation into developments in the relation between pupils' self-confidence, effort, and abilities are planned.

As has been established previously in secondary and higher education, a decrease in school investment over the education period is found here from the beginning of elementary education. This confirms our expectation of decreasing school investment over a school period: decreasing motivation for school, which only improves at the end, before a new start in a different tier of education. This increase in motivation at the end has been found before (Gottfried, Fleming, & Gottfried, 2001). These results suggest that it would be wise to make a school system with more rounded periods. These periods in education should be arranged in a way that students perceive them as a new step in their education again and again. A new step, which will arouse their curiosity, is found to be an important stimulator of intrinsic motivation (Malone & Lepper, 1987).

Although some of individual differences in language development could be attributed to differences in intelligence, there remain differences that can be attributed to pupils' school investment and self-confidence. This confirms the importance of reinforcement of motivation and self-confidence. van der Leij (1985) has already indicated this for pupils with special educational needs, but apparently, it may be equally important for other pupils. Since the development of motivation and self-confidence seems to be relatively unrelated, attention needs to be paid to both. One possibility may be by the use of more formative instead of summative assessments in schools.

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Appendix A. Estimated covariance matrix and means vector

	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.	18.	19.	20.	21.	22.
1. self21	0.714																					
2. self22	0.418	0.785																				
3. self41	0.150	0.126	0.584																			
4. self42	0.122	0.152	0.372	0.722																		
5. self61	0.094	0.081	0.117	0.115	0.601																	
6. self62	0.094	0.116	0.099	0.161	0.420	0.777																
7. self81	0.095	0.095	0.120	0.162	0.121	0.186	0.734															
8. self82	0.071	0.110	0.071	0.178	0.138	0.254	0.505	0.881														
9. inv21	0.079	0.125	0.003	0.020	0.028	0.036	0.063	0.074	0.926													
10. inv22	0.029	0.090	0.027	0.029	0.038	0.056	0.081	0.069	0.578	0.851												
11. inv431	0.014	0.030	0.077	0.088	0.037	0.059	0.032	0.054	0.240	0.231	0.920											
12. inv42	0.011	0.023	0.067	0.080	0.074	0.093	0.078	0.095	0.281	0.278	0.597	1.075										
13. inv61	−0.044	−0.027	−0.022	−0.006	0.086	0.116	0.035	0.052	0.237	0.274	0.322	0.367	1.067									
14. inv62	−0.029	−0.016	0.027	0.030	0.118	0.129	0.086	0.078	0.261	0.300	0.364	0.467	0.664	1.088								
15. inv81	−0.025	−0.035	−0.024	0.017	0.039	0.065	0.141	0.140	0.205	0.222	0.279	0.345	0.383	0.400	1.104							
16. inv82	−0.035	−0.008	0.031	0.061	0.077	0.092	0.155	0.156	0.228	0.269	0.292	0.412	0.379	0.491	0.692	1.078						
17. lan2	0.077	0.055	0.090	0.082	0.078	0.074	0.051	0.022	0.170	0.160	0.150	0.173	0.148	0.140	0.182	0.148	1.221					
18. lan4	0.042	0.010	0.055	0.041	0.068	0.090	0.109	0.096	0.161	0.195	0.178	0.219	0.211	0.210	0.197	0.174	0.563	1.511				
19. lan6	0.053	0.075	0.064	0.064	0.109	0.124	0.149	0.120	0.224	0.209	0.210	0.269	0.265	0.284	0.275	0.310	0.553	0.757	1.405			
20. lan8	0.016	0.037	0.063	0.035	0.113	0.125	0.190	0.150	0.212	0.228	0.212	0.253	0.257	0.293	0.307	0.316	0.537	0.695	0.853	1.320		
21. Figure	0.108	0.113	0.087	0.073	0.230	0.236	0.320	0.247	0.446	0.412	0.298	0.431	0.384	0.501	0.443	0.506	0.539	0.722	0.924	0.865	7.518	
22. Excl.	0.125	0.058	0.118	0.133	0.174	0.206	0.296	0.265	0.425	0.489	0.327	0.474	0.568	0.568	0.486	0.557	0.747	0.862	1.027	0.920	2.984	6.910
Means	3.87	3.58	3.81	3.59	3.83	3.61	3.82	3.59	0.358	3.62	3.53	3.12	3.44	3.21	3.45	3.25	32.43	34.71	36.02	37.25	14.43	11.67

Care should be taken in replication of the current analyses by means of these covariances and means, since they represent the estimated moments provided by M*plus* as a by-product of the FIML estimation method. FIML estimation requires raw data, which are available upon request from the first author.

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